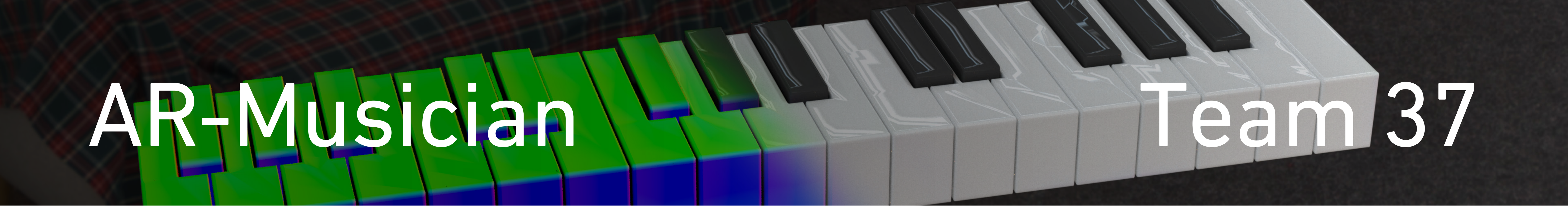


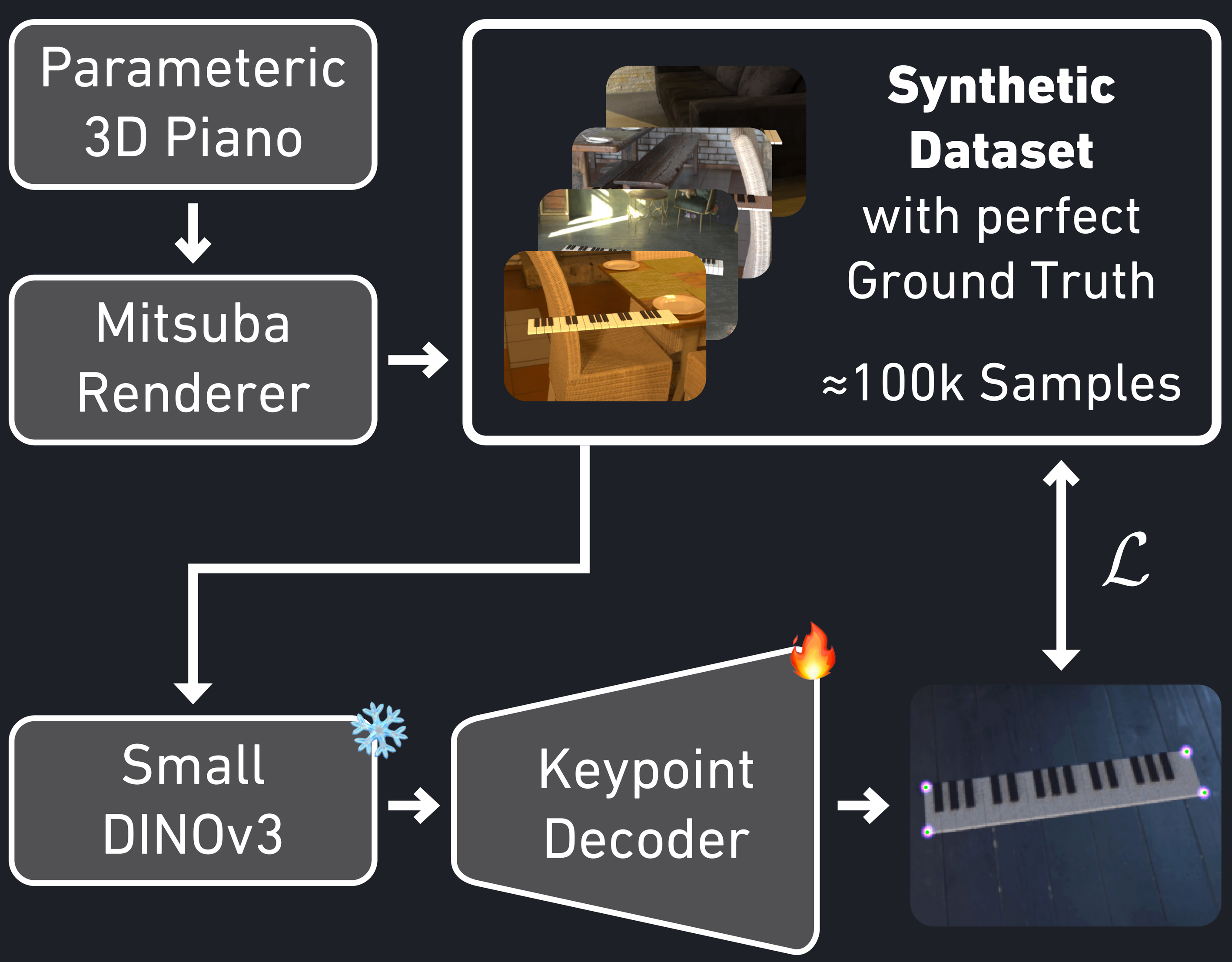
AR-Musician

Team 37

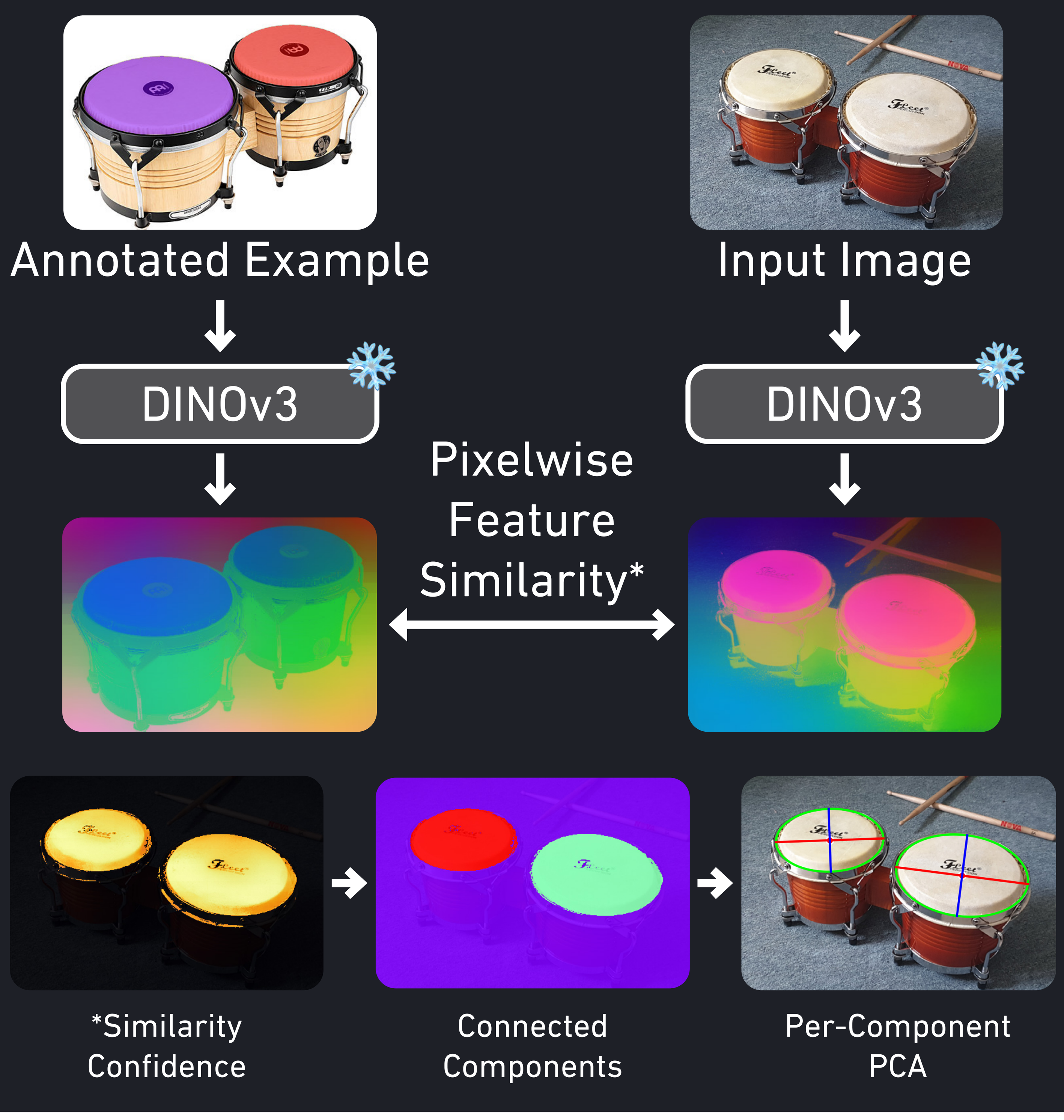
Introduction

Learning a musical instrument presents a significant cognitive challenge, often exacerbated by the inaccessible and distracting nature of traditional sheet music. Mixed Reality offers a powerful solution to this friction by superimposing visual cues directly onto the physical interface, thereby unifying the notation and the instrument. By leveraging computer vision to automatically detect and register the instrument in 3D space, this approach not only simplifies the setup process but also significantly lowers the barrier to entry for aspiring musicians.

Piano Detection



Bongo Detection



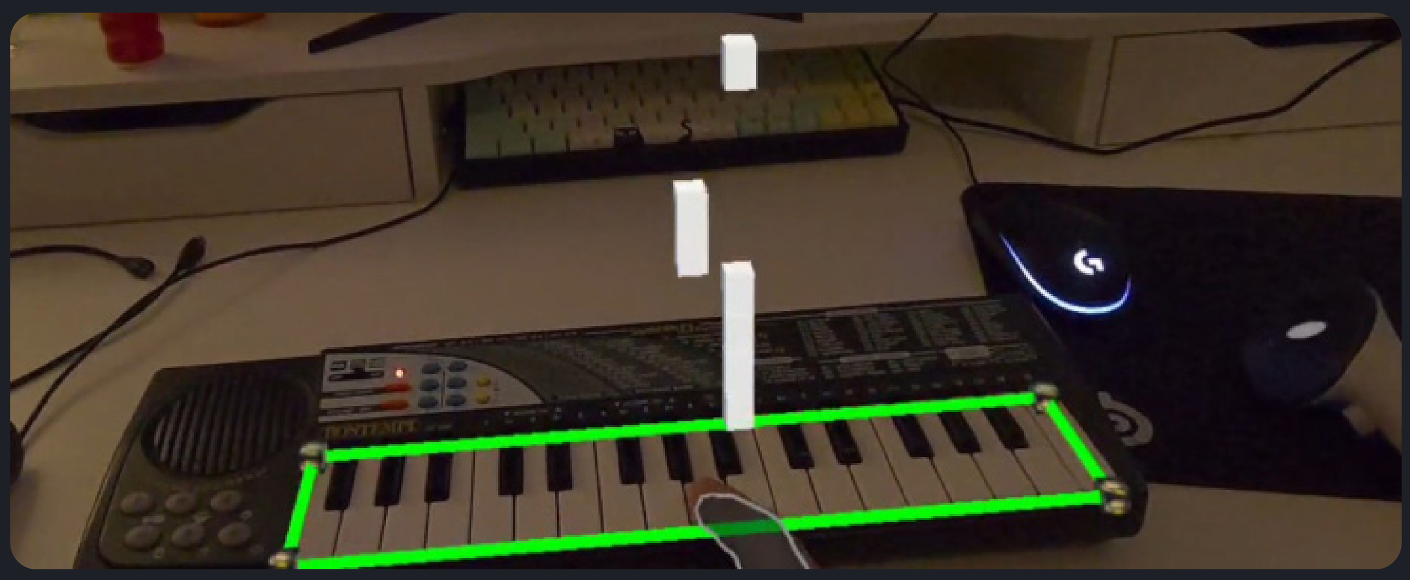
The App

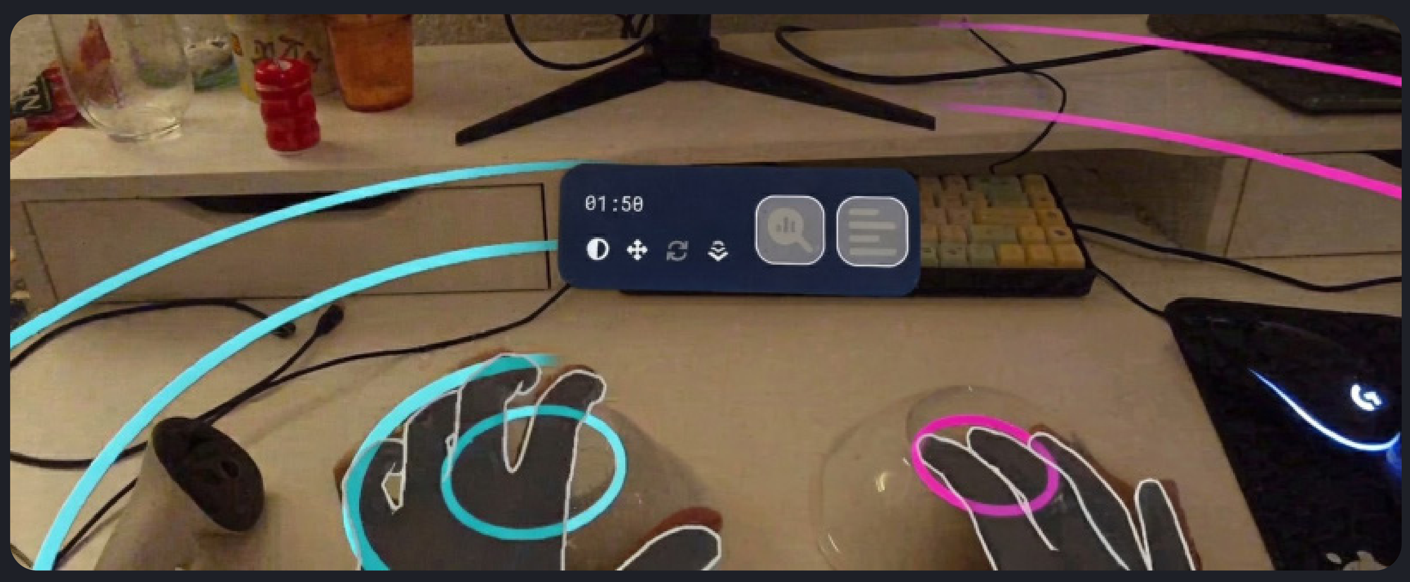
Our application bridges the gap between computer vision and mixed reality interaction through a high-performance pipeline built in Unity:

On-Device Inference Utilizing the Unity AI Inference Engine, our detection models run entirely locally on the Meta Quest. This avoids cloud latency, which is critical for an immersive user experience.

Spatial Registration Pipeline We access the visual feed via the Passthrough API (specifically the left eye camera). Once the ML model parses keypoints, we convert them into 3D rays using camera intrinsics. These rays are projected into the physical room using Meta's Raycasting API to accurately anchor virtual overlays onto physical instruments.

Collaborative Play Built on Photon, the app supports multiplayer sessions. Users can synchronize their practice via custom room keys, enabling real-time remote jamming.





Limitations & Conclusion

Limitations Running a robust object detection model concurrently with a high-fidelity rendering pipeline imposes significant computational overhead on standalone Mixed Reality hardware. To mitigate performance bottlenecks, we ask the users to configure the instrument detection prior to the practice session. This locks the position of the instrument.

Conclusion We have successfully implemented a fully standalone MR instrument tutor that eliminates the friction of manual setup. By training our detection models entirely on synthetic data, we successfully circumvented the prohibitive cost of manual dataset labeling. Future iterations of this work will extend support to a wider variety of instruments and integrate real-time audio analysis, enabling multimodal feedback on pitch and timing accuracy.

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Mixed Reality 2025, ETH Zürich

AR MUSICIAN

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